Did England Deserve to Win? A Data Dashboard Analysis of UEFA Women's Euro 2025

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2026-05-12

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Live dashboard: https://jbuergler.github.io/football-analytics/

Repository: https://github.com/jbuergler/football-analytics

1 Introduction and Research Question

The UEFA Women's European Championship (Euro) is a key international tournament in women's football. In 2025 it was held in Switzerland from 2nd to 27th July 2025. It came at a time when women's football continues to grow in visibility, commercial value, and public attention. The tournament has attracted record television audiences in the United Kingdom (Women's Sport Trust, 2026). Sixteen nations competed across the group stage and knockout rounds, producing 106 goals across 31 matches and setting multiple tournament records (Salley, 2025).

Research on women's football shows that the sport is continuing to grow and is part of it becoming more professionalised. However, there still remain major gaps between the women's and men's game in areas like commercial investment, media coverage, and resources (Tjønndal *et al.*, 2024). That gap is where event data analysis comes in, as it can reveal whether the results are reflected in teams' performances. Anyone who is watching the game, whether it is a fan who is curious about the result, a coach who wants to plan for future fixtures or a journalist writing the post-tournament story, might ask if England's win matched with how they played.

The final match was held at St. Jakob Park on the 27th July 2025, where England competed against Spain. Spain were the reigning World Cup holders, while England entered the title as defending European Champions. Both player of the tournament and top scorer awards went to the Spanish players Aitana Bonmatí and Esther González, respectively (Schwager-Patel, 2025). England retained their European title with a 3-1 penalty shootout win after a 1-1 draw after extra time (Sanders, 2025).

The result raised an immediate analytical question: did it reflect the balance of the play throughout the tournament and in the final itself? In this project, StatsBomb open event data is used to compare England and Spain's performances in the tournament and the final. Metrics like expected goals (xG), pressing, and ball progression are analysed and lead to the main research question:

Did England deserve to win the Women's Euro 2025?

This question matters because although women's football is becoming increasingly more professionalised, there are still major gaps compared to men regarding its visibility, financial conditions, and commercial value (Tjønndal *et al.*, 2024). Recent research also shows that professional women's football has evolved technically and tactically, with event data revealing longer possession sequences, fewer long-distance shots, and more difficult, valuable passes (Bransen and Davis, 2024). This makes event data a useful basis for assessing whether the result reflected the balance of play. For that reason, this project evaluates whether England's title win in 2025 was deserved on performance as well as on the scoreboard.

2 Data Collection and Preparation

2.1 Source and Access

For this project, event data for the UEFA Women's EURO 2025 was sourced via the `StatsBombR` package in R (Hudl, no date). Recently, StatsBomb was incorporated into Hudl's industry-leading ecosystem of integrated video and analytic solutions (Hudl Statsbomb, 2024). They are an established provider of football event data, recording over 3,400 events per match across more than 190 competitions and thus providing a wide range of data free of charge (Hudl Statsbomb, no date). Key functions, including `allclean()` for enriching event coordinates and pass end locations, followed the workflow outlined in the Working with R guide (StatsBomb, 2022, p.25).

2.2 Dataset Structure and Granularity

The dataset consists of event-level data, where each row represents a single action during a match. It includes actions such as passes, carries, shots, pressures, and ball receipts. Each event has information on the player, team, match, time, pitch location (x, y coordinates), and action-specific attributes such as shot outcome and xG value for shots. The x and y coordinates are according to the StatsBomb (2019, p. 34) coordinate system. The pitch goes from 0 to 120 on the x-axis and 0 to 80 on the y-axis. Key variables used in this analysis include:

- **location_x, location_y:** pitch coordinates of the action
- **shot.statsbomb.xg:** expected goals value for each shot
- **shot.outcome.name:** outcome of the shot (goal, saved, off target, etc.)
- **type.name:** event type (shot, pass, carry, pressure, etc.)
- **team.name:** the team performing the action
- **player.name:** the player performing the action
- **minute, period:** time context of the action

2.3 Cleaning and Preparation

In order to prepare the data for the analysis, several cleaning steps were taken:

Period filter: StatsBomb records penalty shootouts through the period variable 1-5, with the 5 assigned to the shootout. This is based on single shots and not what happened in open play. Therefore, all period 5 events were excluded from the analysis. A total of three matches were impacted by this, including the semi-final between Sweden and England, the quarter-final between France and Germany, and the final between England and Spain.

Own goals: As an own goal does not reflect the performance of the attacking team, they were excluded from all shot and xG analyses. There were a total of 3 own goals, which makes the analytical goal count (103) different from the 106 goals shown on the dashboard.

Team name standardisation: To ensure consistent and clean team names across all tables and visualisations, all prefixes and suffixes like “Women’s” or “WNT” (for Finland) were removed.

Pre-aggregation: For all metrics used in this project, summary tables were created first and saved as `.rds` files. They were loaded directly by the Shiny dashboard. This approach improves loading speed and separates the data preparation from the app itself.

The cleaning process produced two output tables: `weuro2025_events_clean.rds`, which includes all event types and is used for pressing and possession analysis. `weuro2025_shots_clean.rds` contains shots only and is used for all XG and shot analysis.

2.4 Coverage and Caveats

As shown in Table 1, the dataset covers all 31 matches of the UEFA Women’s Euro 2025 tournament from the group stage through to the final. As coverage is subject to StatsBomb’s data collection methodology, any recording errors cannot be ruled out.

This project analysed on-ball actions only. Off-ball movements and physical metrics such as distance covered are not included. This means that some aspects of performance, especially defensive organisation away from the ball, were not covered in this analysis. StatsBomb's freeze frame data, which records the location of all players involved in the event, including goalkeeper and defender positions at the moment of each shot, was not used in this analysis (Hudl Statsbomb, 2021). However, it informed the pre-computed xG values available in the dataset, which makes them more contextually accurate. It should be noted that these xG values reflect StatsBomb's model assumptions and cannot be independently verified from the open dataset alone, but remain a well-established and widely used metric in football analytics (Matcham, no date). The dataset recorded a total of 105,525 individual events, including 875 shots (excluding own goals and penalty shootouts).

StatsBomb records concrete pressing events when a player applies pressure to an opponent in possession, instead of computing pressure through combining multiple defensive actions (Dorrington, 2025). This is an advantage, as it allows pressing intensity to be evaluated directly from recorded events.

Table 1: Dataset Summary - UEFA Women's Euro 2025

Item	Detail
Source	StatsBomb open event data
Access method	StatsBomb R package
Competition	UEFA Women's Euro 2025 (ID: 53, Season: 315)
Matches	31
Shots	875
Granularity	Event level (one row per action), a total of 105,525 events
Period 5 excl.	Yes (penalty shootouts)
Own goals excl.	Yes (from xG and shot analysis)

3 Methodology

3.1 From Raw Data to Dashboard

This project follows a structured pipeline of five scripts, moving from raw data collection through to the interactive dashboard. First, raw event data was pulled from StatsBomb in script `01_data.R`. Afterwards, the data was explored (`02_explore.R`) and cleaned (`03_clean.R`). Following that, the data was aggregated into summary tables (`04_analyse.R`). The plots were first created as static visuals in `05_visualise.R`. Once the static versions were confirmed, they were moved as interactive elements in `app/app.R`. The visualisation process was iterative, with some debugging and changes made based on the app output on Shinylive. Some plots had to be converted to static elements instead due to compatibility issues. Once the app ran successfully, the dashboard was then deployed to GitHub. These steps ensure that the analysis can be reproduced and the dashboard only uses smaller, cleaned data tables that are easier to load through Shiny.

3.2 Metrics

All metrics were calculated based on the cleaned event data and consolidated into summary tables. The following metrics were used across the dashboard:

Expected goals difference (xG diff): The xG difference was calculated as a team's total minus their opponent's xG. The metric was then averaged across all matches for a tournament-wide measure. It illustrates whether a team consistently created better chances than they conceded. This is the primary metric for the Tournament Overview tab of the Dashboard.

High press rate: High press rate measures how often a team applies pressure in the opponent's half. Any pressure event that happened beyond `location_x > 60` on the StatsBomb coordinate system counted as a high press. StatsBomb records a pressure event each time a player actively applies pressure to an opponent in possession, which is a direct measure of pressing behaviour.

Counter-press rate: When a team loses the ball and immediately tries to win it back, StatsBomb flags this as a counterpress. The counter-press rate is the share of all pressure events that were classified as such. Missing values were recoded to `FALSE` so the full pressure count could be used to compute the rate in %.

Pass completion rate: The proportion of attempted passes that successfully reached a teammate from all open play passes. Unknown outcomes and injury clearances from `pass.outcome.name` were excluded. It is worth noting that in the StatsBomb data, `NA` values in `pass.outcome.name` refer to completed passes (StatsBomb, 2019, p.28).

Final third carry rate: This refers to the proportion of all carries ending beyond `x = 80` on the StatsBomb coordinate system, which represents the final third of the pitch. This metric was chosen over the standard StatsBomb progressive carry definition (carries advancing the ball by at least 10 units). As it directly shows penetration into the attacking third instead of forward progress from any location, this was chosen as a more meaningful metric to analyse the attacking play.

Final third pass rate: Completed passes that ended beyond `x = 80`were counted as final third passes. The rate is the share of all completed passes that reached this zone. It shows how often each team attempted to move the ball into the attacking third.

Shots faced per match: This is the average number of shots conceded per match and used to measure defensive solidity, together with the average xG conceded.

3.3 Grouping and Filtering Decisions

All pressing metrics were split by stage type (group stage vs knockout rounds) to capture if there were any tactical shifts as the tournament progressed. The final match was isolated by selecting only the match ID of the final itself using `competition_stage == "Final"` and analysed separately across the tabs that focused on a detailed analysis of the events in the final between England and Spain. Actions of the best players were filtered to Bonmatí and Hemp specifically. They accumulated the most final third passes and carries for their teams. The pass map visualisation was made based on the approach described in the Working with R guide by StatsBomb (2022, pp. 22-24).

3.4 Dashboard Design

The dashboard was built using R Shiny with the `bslib` package for the layout and theme (Sievert, 2023). A consistent colour palette was applied for the entire dashboard based on the official hex colours of Spain in gold (`#F1BF00`) and England in red (`#CE1124`). All other teams were highlighted in grey to ensure England and Spain can be immediately identified across every chart.

The dashboard was structured across a total of six tabs that guide the reader through the story of the tournament and ultimately answer the research question. Each tab also includes a navigation sidebar that can be collapsed if needed. This gives some initial context of what each tab is showing.

In the first tab, Tournament Facts and Context was provided, as well as manager, formation, and high-performing player information for England and Spain. After that, the Tournament Overview provided context for each team's performance regarding their expected goals (scored and conceded). Moving on, the Journey Tab shows the development of the two finalists throughout the tournament across metrics like accumulated xG, match-by-match xG and pressing. Afterwards, the final was analysed in more detail across two tabs to analyse the shot and xG situation, as well as the Possession in separate windows. The final was shown across two tabs on purpose as the plots would have looked too cluttered and difficult to read. In the last tab, the research question was answered using the analysed metrics.

The individual chart types were chosen based on the analytical question each visualisation attempted to address. A horizontal bar chart (Tab 1) ranks all 16 teams by xG dominance, making the ranking immediately readable. A scatter plot (Tab 1) compares total xG against goals scored, showing which teams over- or underperformed in terms of goals. Cumulative line charts show how the xG and momentum change across the tournament, while step charts show the xG accumulated shot by shot in the final. Pitch-based plots like shot maps (scatter plot) and player passmaps (arrow maps) use spatial data to show where on the pitch certain actions happened. Reference lines at +1 and −1 are added to the xG ranking chart to provide context for what a meaningfully positive or negative xG difference per match refers to.

Apart from two exceptions, all plotly charts include hover tooltips providing exact values for each element of the plots. The xG Dominance ranking chart and player passmaps use static `renderPlot` instead of `plotly`. For the passmaps, this was due to a compatibility constraint between the `ggsoccer` package and the rendering mechanism of `plotly` in the Shinylive WebAssembly environment. For the xG ranking chart, the static approach was chosen because of a Shinylive WebAssembly issue with categorical variables on the y-axis. This caused issues in rendering the plot with `plotly`. This was solved by converting the plot to a static one, which is acceptable, as interactivity does not add meaningful value to a ranked chart. The relative positions and reference lines at ±1 communicate the findings clearly without tooltips.

4 Findings

4.1 Tournament Overview

Figure 1 shows that England and Spain outperformed relative to their expected goals. Spain scored 18 goals from 15.91 xG (+2.09), and England scored 16 goals from 14.68 xG (+1.32). Both teams were clinical in front of goal, with Spain being slightly more clinical and also generating more chance quality.

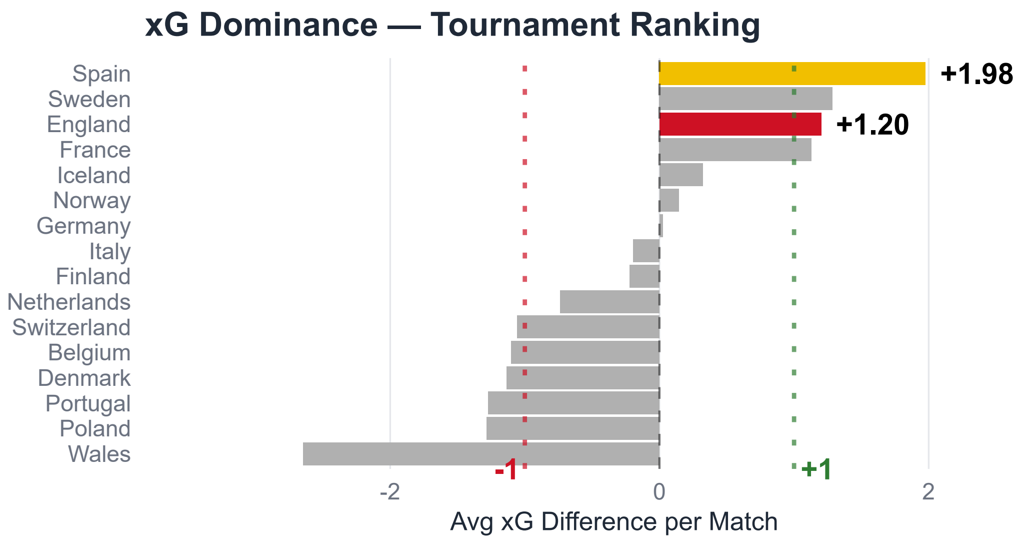

Figure 1: All 16 Teams ranked by average xG Difference per Match. Spain led the Tournament (+1.98), ahead of England (+1.20).

In terms of average xG difference, Figure 2 illustrates how Spain outperformed all teams with an average xG difference of +1.98 per match. That means they created significantly better chances than they conceded across all their games. England ranked third with an average xG difference of +1.20 per match. This reveals that both finalists performed above the +1 reference line by creating more and better chances than they allowed. Wales had the lowest average xG difference per match (-2.65), which confirms their early group stage exit. However, this number is slightly inflated due to their 1-6 loss against England on Matchday 3.

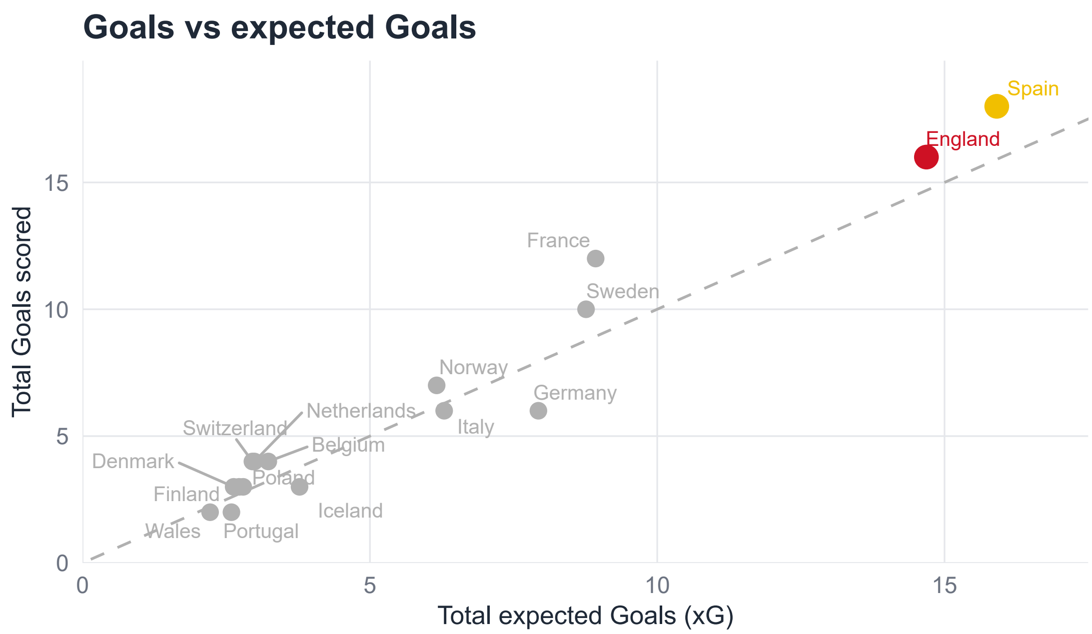

Figure 2: Goals scored vs total xG for all 16 Teams. Teams above the dashed line scored more than expected, and teams below the dashed line scored less than expected. Both England and Spain outperformed their xG, with Spain generating greater overall chance quality.

4.2 The Journey to the Final

To compare England's and Spain's routes to the final and how they performed one by one until playing each other in Basel, several metrics were analysed.

Figure 3 depicts how both teams' cumulative xG increased over the whole tournament. It shows that Spain's xG increased consistently while England's increase was influenced by their 6-1 win against Wales on Matchday 3. Over the rest of the tournament, both teams' values were at ~ 13.80 after the semi-finals. After the final game on July 27th, Spain clearly took the lead over England, having an overall difference of +1.23. Spain had a more consistent attacking output throughout the tournament compared to England.

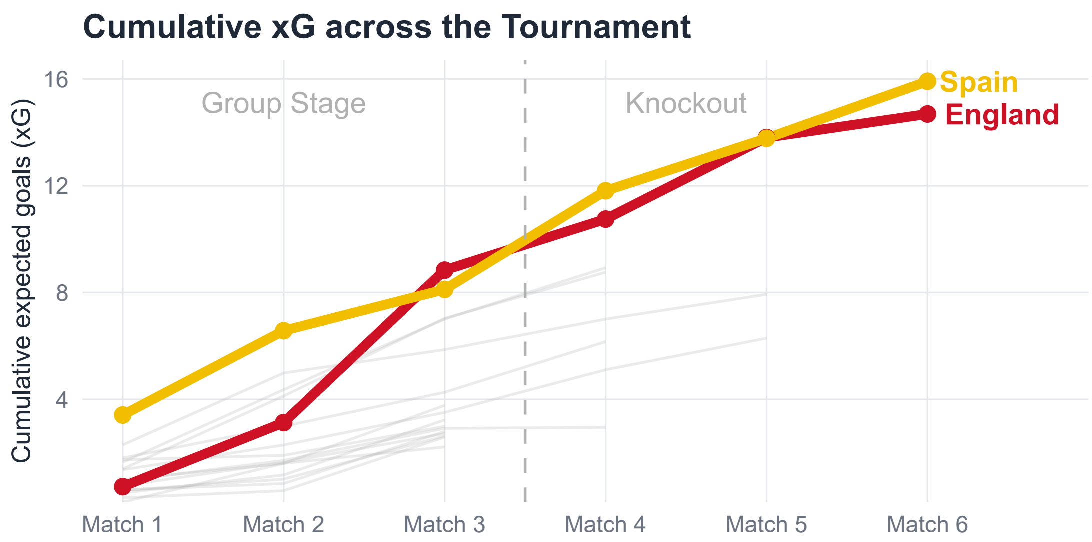

Figure 3: Cumulative xG across the tournament for England and Spain. Spain built xG at a consistently higher rate, pulling clear from Match 4 onwards. Grey lines show all other teams.

Figure 4 and Figure 5 further confirm the cumulative xG trend. They show the xG breakdown for each of England's and Spain's games compared with their opponents. As already seen in Figure 3, England's xG was largely shaped by their massive win over Wales, where they achieved their highest single-match xG of the tournament (5.71). If that outlier was removed, England's attack looked a lot more modest. Spain, however, confirm the previous trend of having a more consistent xG pattern. They rarely generated less than 2 xG per game, no matter who they played.

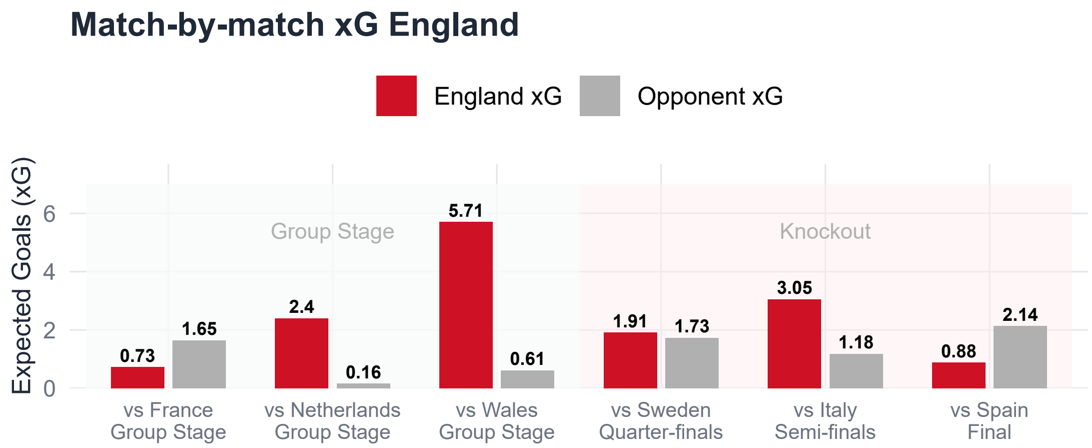

Figure 4: Match-by-match xG for England across the Tournament. Red bars show England xG, grey bars show opponent xG. England's group stage total is heavily influenced by the 6-1 win against Wales.

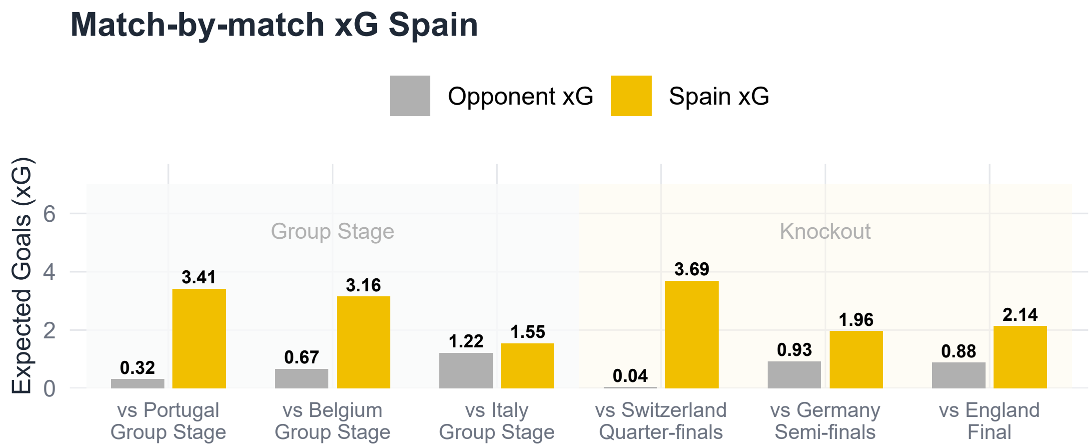

Figure 5: Match-by-match xG for Spain across the Tournament. Yellow bars show Spain xG, grey bars show opponent xG. Spain maintained consistent attacking output throughout.

Figure 6 shows the pressing intensity across high pressing and counter pressing. Evaluating both these metrics, Spain was ahead of England in the group stages and knockouts. They achieved their highest rates in the group stages with 77% (high-press rate) and 32% (counter-press rate). Spain's numbers suggest a team that set up to win back the ball high up the pitch and keep opponents in their own half. This style fits with their xG advantage across the tournament. The data, however, shows that Spain, in particular, reduced their pressing intensity in the knockouts, compared to the group stages. This is not surprising, considering the margins to win in the knockouts are usually higher, and a pressing done wrong could lead to a fatal counterattack. Surprisingly, England's counterpressing only marginally reduced in the later stages of the tournament (-2.50%). This shows that there was a smaller tactical change compared to Spain.

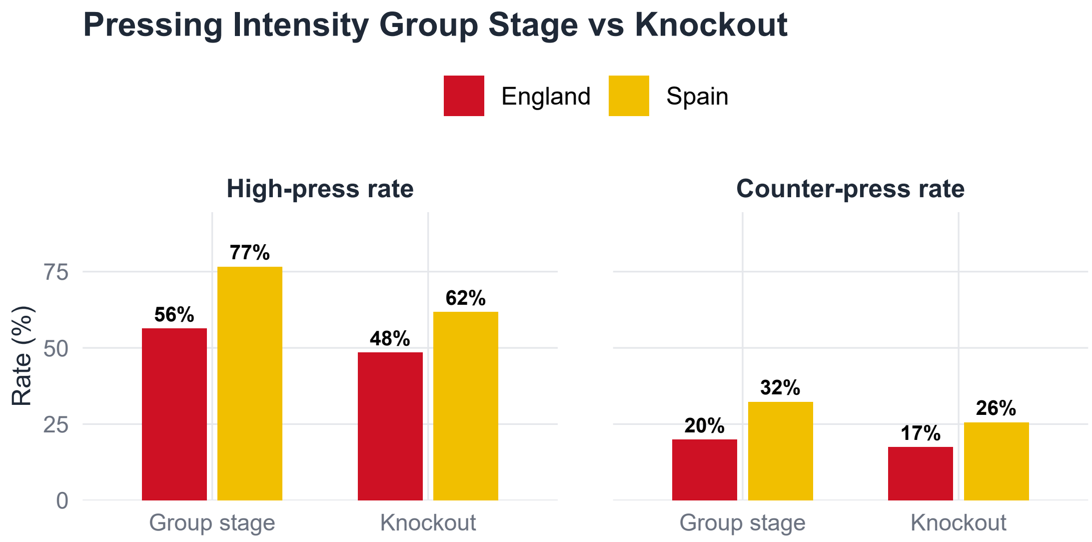

Figure 6: Pressing Intensity by stage. High press rate and counter-press rate for England and Spain across the group stage and knockout rounds. Spain pressed higher at every stage.

4.3 The Final

Looking at the attacking output in the final, Spain clearly dominated England on both analysed metrics. Figure 7 shows Spain had 23 shots compared to England's 8, a volume that shows their attacking dominance. The location of Spain's attempts was also more centrally located around the penalty area, compared to England's more dispersed shot ranges. Both goals in the match were headers. Spain's Caldentey headed in from close range in the 24th minute (0.396 xG), while England's Russo equalised with a header in the 56th minute (0.213 xG).

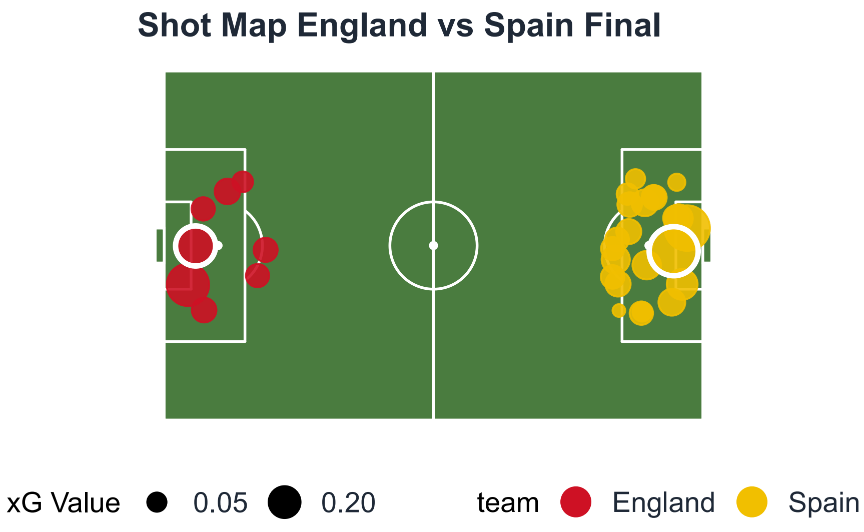

Figure 7: Shot map for the Women's Euro 2025 Final. England attack from the left, Spain attack from the right. Circle size represents xG value. The white ring indicates a goal. Spain registered 23 shots to England's 8.

Figure 8 further illustrates how Spain dominated in attack from the start of the match. They accumulated a total of 2.14xG across their 23 shots by the end of extra time. England only had a total xG of 0.88 from 8 shots, but had a higher xG per shot (0.11 xG vs 0.09 xG). England's last shot came in the 68th minute, which means they did not create any chances in the final 52 minutes of the game, including extra time. This does not necessarily mean a lack of ideas. It suggests an intentional second-half adjustment. After Russo equalised, England stopped trying to actively win the game through open play attacks and focused on keeping the score level. The match ended 1-1 after 120 minutes despite this xG gap and Spain's more dominant attacking display. England then won the game 3-1 on penalties, which contradicts the attacking performance of the game.

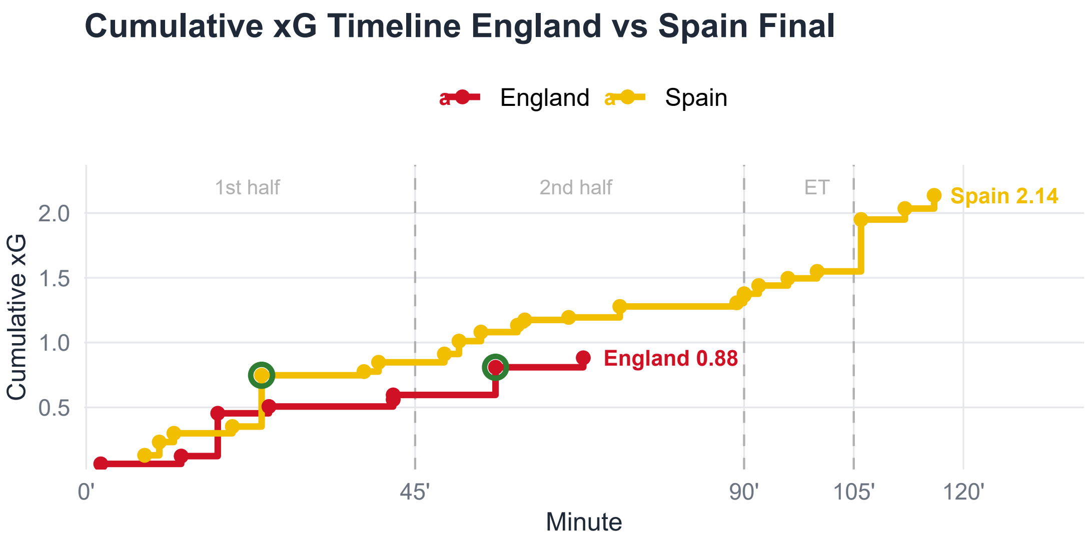

Figure 8: Cumulative xG Timeline for the Women's Euro 2025 Final. Spain dominated throughout, finishing with 2.14 xG. England's last shot was in the 68th minute. Green rings indicate goals.

Spain also dominated possession. As shown in Figure 9, they directed 29% of completed passes into the attacking third, compared to England's 15%. It further displays that England made more passes in defensive areas of the pitch (37% vs 22%). Across both shots and possession, it is clear that Spain attempted more shots and ball movements into the final third. England's more defensive dominance implies they had a more conservative and compact approach in the final.

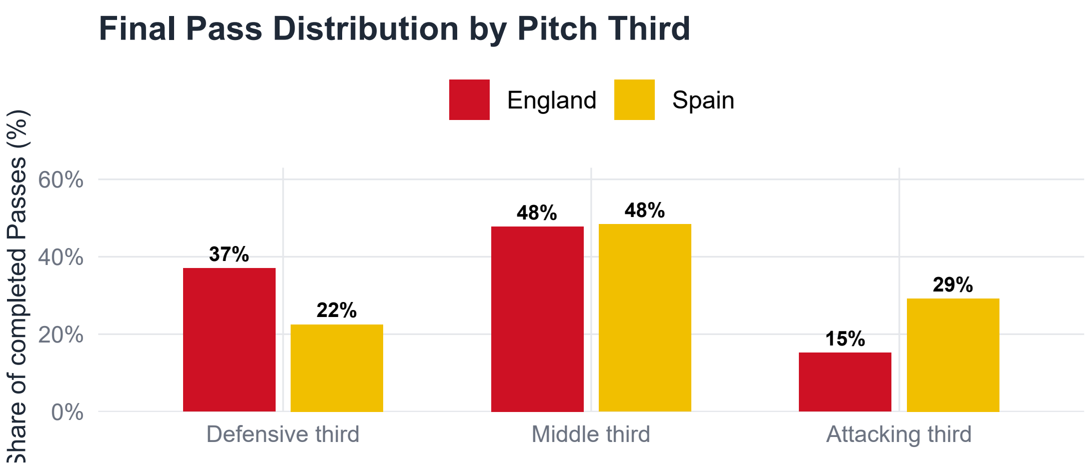

Figure 9: Completed pass distribution by pitch third in the final. Spain directed more passes into the attacking third, while England was more concentrated in the defensive third.

In the final, Bonmatí and Hemp showed clear differences in their final-third actions. Despite both players playing the full 120 minutes, Bonmatí created much more attacking threat. As seen in Figure 10, Bonmatí played in the entire attacking third, making passes and carries to the final third from central and wide areas. On the other hand, most of Hemp's actions were concentrated around the left side. This matches England's more conservative attacking style.

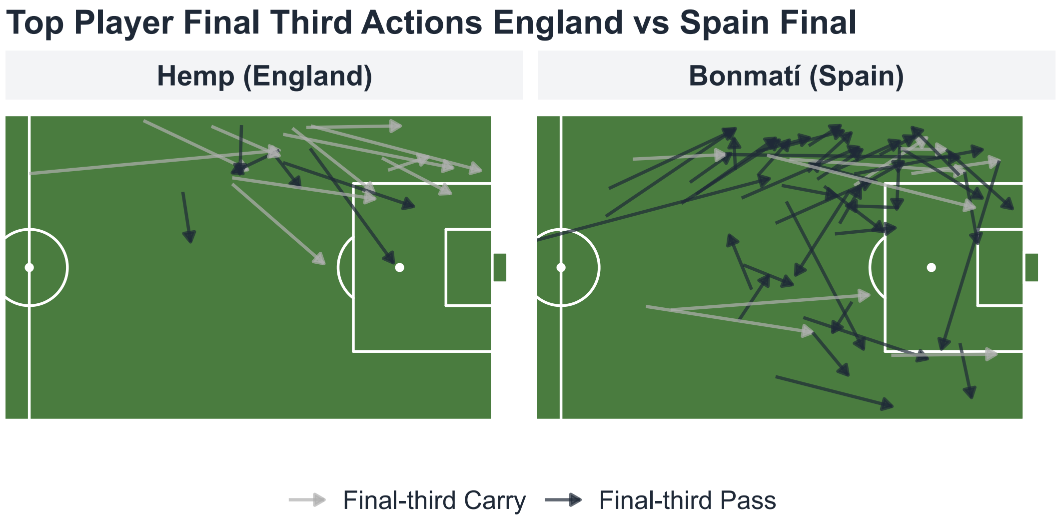

Figure 10: Final-third actions for Bonmatí (Spain) and Hemp (England) in the final. Arrows show completed passes and carries ending in the final third. Bonmatí operated across the full width of the attacking third while Hemp was concentrated through the left channel.

4.4 Analytical Outcome

As shown in Table 2, Spain led England on all seven performance metrics across the tournament. Spain averaged more xG per match (2.65 vs 2.45) and conceded less as well (0.67 vs 1.25 xG per match). They also dominated in terms of pass completion (87.2% vs 79.4%). This number alone does not indicate better possession. But Spain showed their attacking threat through other metrics. Particularly, for the final third carries, Spain had a 6.6 percentage point lead compared to England. Consequently, by attacking more, this also led to fewer shots faced per match (8.7 vs 11.8).

One finding that complicates this picture is England's pressing in the final. Spain had a much higher high-press rate across the tournament (67.8% vs 51.5%). But England applied 393 pressures to Spain's 253 in the final. This suggests that England's tactical approach was more offensive than the xG gap would indicate. Their defensive resilience was a big part of that. The data, however, showed Spain was the better side in all seven of the tournament-wide metrics.

Table 2: Key performance metrics — England vs Spain across the tournament. Spain led on all seven metrics.

Dimension	Metric	England	Spain
Chance Quality	Avg xG per match	2.45	**2.65**
Defensive Solidity	Avg xG conceded per match	1.25	**0.67**
Game Control	High press rate (%)	51.50	**67.8**
Game Control	Pass completion (%)	79.40	**87.2**
Attacking Threat	Final third carry rate (%)	29.10	**35.7**
Attacking Threat	Final third pass rate (%)	34.70	**37.9**
Defensive Solidity	Shots faced per match	11.80	**8.7**

5 Discussion, Limitations, and Conclusion

5.1 Implication of Findings

The numbers are quite clear and show that Spain was the stronger side. They created more and better chances (2.65 vs 2.45 xG per match), conceded far less (0.67 vs 1.25 xG against), and pressed higher at every stage of the tournament, as shown in Table 2. The gap was even wider in the final, where Spain created 2.14 xG from 23 shots compared to England's 0.88 from 8. In terms of attack and defence, Spain should have won.

It is the way England won that makes the result worth looking at. Spain had pressed higher than England at every stage of the tournament. In the final, England were the more aggressive team with 393 pressings to Spain's 253, which is the opposite of the other games in the tournament. Combining this with a pass share mainly focused in their own defensive third suggests a team that chose to absorb pressure instead of attacking themselves. By being defensively strong, they were able to limit Spain's space and get the game to penalties.

The result shows what is regularly observed in football, where match outcomes are heavily influenced by randomness and may not always reflect underlying team performance (Brechot and Flepp, 2020).

The data has a different use depending on who is looking at it. If a coach is looking to prepare for a future fixture against Spain, they could look at how their pressing dropped between different stages of the tournament (from 77% in the group stage to 62% in the knockouts). That decrease suggests Spain presses less aggressively as the stakes in a tournament increase. This is worth considering when planning how to build attacks against them in a knockout game. For a journalist, the xG timeline of the final is the clearest story. Spain dominated for 120 minutes, England's last shot was in the 68th minute, and penalties decided a match that statistics alone cannot fully explain.

5.2 Limitations

Three limitations should be acknowledged before concluding this analysis. First, this analysis only focused on on-ball event data. Any movements off the ball, such as defensive shape and physical metrics like distance covered or sprint intensity, were not covered and therefore outside the scope of this analysis. Especially England's defensive organisation in the final cannot be fully quantified through the metrics used in this analysis.

Second, small sample sizes have an impact on the metrics. The maximum number of matches a team could play was six, and this only if the team went through to the final. The majority of the teams only played three games (group stage), or potentially one or two extra games in the knockout stages. For example, England's xG figures are visibly inflated by their 6-1 win against Wales in the group stage, and vice versa, Wales' average xG difference (-2.65) is disproportionately negative for the same reason. By getting the average across all matches, some useful comparisons can be drawn. But they should be interpreted with caution due to the sample size and the number of games each team played.

Third, the player-level analysis only focused on Bonmatí and Hemp, who were the players with the most final-third actions for their respective teams. Although this was a deliberate decision, it does not capture other important contributions. For instance, England Goalkeeper Hannah Hampton was named player of the match in the final (UEFA, 2025). This suggests she played a key role in England's defence and the penalty shootout to help them retain their European title.

5.3 Future Outlook

There are several extensions to this analysis to further strengthen it. For example, the StatsBomb freeze frame data could be included. This could help evaluate shot quality in a more accurate way by including goalkeeper positioning and defensive pressure at the moment of each attempt. Second, if available, physical tracking data should be incorporated. The analysis could be improved by, for example, quantifying England's defensive shape instead of implying it. The same analysis could also be extended to other international tournaments, like the 2022 Women's Euro or the 2023 World Cup. This could help to put England's and Spain's 2025 performances into context by comparing them to their own past results. Another way could be to look at the match situation in each game, which shows whether a team was winning, drawing, or losing. This could add useful context and help in highlighting intentional tactical changes in attack and defence. This would be particularly interesting to interpret England's second-half and extra-time pressing in the final to illustrate how they were able to keep the result at 1-1 without attempting to shoot themselves.

Of these, incorporating freeze frame would be the simplest next step with the most direct impact on the research question. It is available through StatsBomb and would allow England's defensive positioning at the moment of each Spain shot to be quantified. This could potentially explain why Spain's 2.14 xG only resulted in one goal. Out of the other additions, full tracking data would have the most comprehensive impact on improving the analysis, but would require access to the respective data, which may be more difficult to achieve.

5.4 Conclusion

Going back to the initial question of whether England deserved to win the Women's Euro 2025, based on performance, the data suggests otherwise. Spain was the stronger side throughout the tournament and in the final, and the metrics analysed in this project support that. England retained their title through tactical discipline, clinical finishing, and penalty shootout success. All that matters a lot in a knockout tournament, where the state of the game can shift from one moment to the other. However, not all of those factors can be fully captured by event data alone.

None of this diminishes what England achieved. In knockout football, resilience counts for as much as control. England showed both when it mattered most. They conceded just once in

120 minutes and converted three of their penalties when it counted. The data helps to explain how the game played out. It does not explain Chloe Kelly stepping up for England in the 120th minute to convert the decisive penalty to secure their win.

List of Tools Used

Anara Technologies (2026) *Anara* (Pro Version). Available at: https://anara.com

- Assisted in understanding and summarising academic literature
- Facilitated the organisation of research findings

Anthropic (2026) *Claude* (Version claude-sonnet-4-6). Available at: https://claude.ai

- Assisted in debugging R code and Shiny app development
- Support with structuring report sections
- Proofreading and grammar check for cohesion and readability

Grammarly Inc. (2026) *Grammarly* (Free Version). Available at: https://www.grammarly.com

- Support with spelling and grammar checks, including punctuation and preposition use

QuillBot (2026) *Paraphrasing Tool* (Free Version). Available at: https://quillbot.com/paraphrasing-tool

- Assisting with paraphrasing sentences and sentence structures
- Finding synonyms

Zotero (2026) *Zotero* (MacOS Version). Available at: https://www.zotero.org/

- Assisting in the organisation of references and citations

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